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Next item →

1. The mean or expected value of a scalar random variable is a special case of the expression for the mean of a vector random variable. If a scalar random variable has pdf $f_X(x)$, then its mean is computed as $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) \, dx$.

1 / 1 point

For this problem, consider a scalar random variable X having the following pdf:

$$f_X(x) = \begin{cases} 1 & 1 < x < 2 \\ 0 & \text{otherwise.} \end{cases}.$$

What is $\mathbb{E}[X]$? Enter your answer with one digit to the right of the decimal point.

1.5

✔ Correct

Yes. That is the correct calculation.

2. The variance of a scalar random variable is a special case of the expression for the covariance of a vector random variable. If a scalar random variable has pdf $f_X(x)$, then its variance is computed as $\mathbb{E}[(X - \bar{x})^2] = \int_{-\infty}^{\infty} (x - \bar{x})^2 f_X(x) \, dx$, where $\bar{x} = \mathbb{E}[X]$.

1 / 1 point

For this problem, consider a scalar random variable X having the following pdf:

$$f_X(x) = \begin{cases} 1/12 & -6 < x < 6 \\ 0 & \text{otherwise.} \end{cases}.$$

What is the variance of random variable X ? Enter your answer with one digit to the right of the decimal point. (Hint: $\mathbb{E}[X] = 0$ for this particular random variable.)

12

✔ Correct

Yes. You have performed the correct calculation.

3. Consider a vector random variable X having multivariable Gaussian pdf with mean

1 / 1 point

$$\bar{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and covariance } \Sigma_{\bar{x}} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Compute $f_X(x)$ for $x = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Enter your answer with 3 digits to the right of the decimal point.

0.053

✔ Correct

Yes. That is the correct computation.